

Ronald Ephraim Tiangson

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Career Summary

Senior Software Engineer with 16+ years of experience designing and delivering scalable enterprise and cloud-native systems across finance, insurance, healthcare, and SaaS industries. Currently completing a Master of Software Engineering in New Zealand, specialising in software architecture, AI systems, and cloud engineering.

Experienced in building API-first platforms using .NET and Python (FastAPI), applying Clean Architecture, SOLID principles, and DevOps best practices. Proven track record of modernising legacy systems, improving performance, and delivering production-ready solutions that reduce operational overhead and increase reliability. Passionate about building ethical, high-impact technology and mentoring high-performing teams.

Skills

- **AI & Data:** Python, FastAPI, LLM Integration, FAISS Vector Search, OCR Pipelines, Data Analysis, Exploratory Data Analysis
 - **Programming:** C#, .NET Core, .NET Framework, VB.Net, VB6, VBA
 - **Mobile & Frontend:** React, Angular, TypeScript, JavaScript, ASP.NET MVC, REST APIs, HTML5, CSS, Bootstrap, Android (Kotlin), PHP
 - **Cloud & DevOps:** AWS (Lambda, API Gateway, RDS), Microsoft Azure, Oracle Cloud, Docker, Kubernetes, GitHub Actions, Azure DevOps, Bamboo, CI/CD Pipelines
 - **Databases:** SQL Server, MySQL, Oracle, SQLite, DB2, MS Access, SSIS, SSRS, Crystal Reports
 - **Testing:** NUnit, xUnit, SpecFlow, Moq, Selenium, Cypress, MS Unit Test
 - **Architecture & Practices:** Clean Architecture, Domain-Driven Design (DDD), DTO–Mapper–Repository Pattern, SOLID Principles, Microservices, Layered Architecture, OOP, SOA
 - **Tools:** Git, GitHub, TFS, SVN, Jira, Confluence, Visual Studio, Azure Data Studio
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Work Experience

Application Analyst Developer

Satellite Office / Smartgroup Australia | Jul 2023 – Jul 2025

- Developed new apps and provided maintenance support for legacy systems in the corporate space.
 - Optimized SQL stored procedures to improve data retrieval speed.
 - Developed reports with SSRS and PDF Pro to provide stakeholders with data-driven insights.
 - Conducted code reviews and enforced coding standards to ensure long-term maintainability.
 - Implemented unit, integration, and automated tests to guarantee system stability.
 - Collaborated with project managers and business analysts to align features with client goals.
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Senior Software Engineer

Willis Towers Watson | Mar 2020 – Jul 2023

- Supported and enhanced employee benefits systems (BrightChoices, Benefit Access, FSA Portal).
 - Designed, developed, and delivered features end-to-end using .NET and C#.
 - Implemented automated testing with Selenium, SpecFlow, and Cypress, reducing defects by **25%**.
 - Facilitated knowledge-sharing sessions to upskill teams.
 - Assisted recruitment efforts by conducting technical interviews and evaluations.
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.Net Core Developer

AXA Philippines | Aug 2018 – Mar 2020

- Developed and implemented new components for ACE Solutions Delivery.
 - Proactively minimized troubleshooting and errors through teamwork and problem-solving.
 - Ensured high reliability by writing **100% unit test coverage**.
 - Enhanced reporting accuracy and performance through SQL optimization.
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Earlier Roles

Selected Academic & Personal Projects

KaiHelper – AI-Powered Grocery Budgeting App *Master of Software Engineering – Team Project*

- Designed and built a **production-ready budgeting platform** using **Python FastAPI**, Clean Architecture, DTO-Mapper-Repository pattern, and SOLID principles.
- Implemented **receipt image parsing (OCR + GPT-4o)** to automatically extract grocery items and expenses.
- Built **budget tracking, categories, funding, and insights** modules with clear domain separation.
- Planned deployment using **AWS (Lambda, API Gateway, RDS)** with cost-conscious architecture.
- Integrated **Android (Kotlin) client** consuming REST APIs.
- Emphasised real-world product thinking: scalability, maintainability, and user-centric design.

RecallAI – Personal Knowledge & LLM Assistant *Independent Project*

- Architected a **self-hosted AI assistant** using **FastAPI**, **local LLMs** (DeepSeek, Qwen, TinyLlama), and **FAISS vector search**.
- Implemented document ingestion, embeddings, semantic search, chat, and multimodal pipelines.
- Deployed on **Oracle Cloud Free Tier VM**, optimising for low-cost, offline-first inference (no paid APIs).

- Designed modular AI services for future expansion (chat, code generation, research, transcription).

Data Analytics, ITS & Quantum Computing (Academic)

- Performed **exploratory and confirmatory data analysis** on public transport datasets (delays, traffic, weather, passenger load).
 - Designed **Intelligent Transportation System (ITS)** recommendations using real-world constraints.
 - Conducted **quantum computing experiments** (Bell states, Grover's algorithm) using Python-based tooling and simulators.
 - Produced research-grade reports aligned with postgraduate assessment standards.
 - **Full-stack Developer – L.R.B. Medical & Maternity Clinic (2018)**
Built and maintained a clinic system using .NET Core, Angular, and SQL Server; developed workflows and reports that improved operational efficiency.
 - **C#.Net Developer – Eclaro International Inc. (2015 – 2017)**
Converted DB2 to SQL Server, re-engineered apps, and enforced coding standards.
 - **Lead Software Analyst – Equitable Computer Services, Inc. (2008 – 2015)**
Managed a 3-developer team and delivered high-volume applications while mentoring junior engineers.
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Education

Master of Software Engineering – In Progress (2025–2026)

Yoobee College of Creative Innovation, New Zealand

Bachelor of Science in Computer Science

Fillamer Christian College, Roxas City, Philippines (2003 – 2007)

References

Available upon request